

Unit 14: Network Security

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Objectives

- understand basic of network security.
- understand network security issues
- learn security goals
- understand security services
- approaches of network security

Introduction

To begin, let's define some fundamental concepts in network security, such as plaintext, ciphertext, encryption, decryption, cryptography, and cryptanalyst as seen in Figure 1. And there's the key. So, what do all of them imply? I'll explain you what they imply in a very easy way so you can comprehend what plaintext ciphertext encryption decryption means.

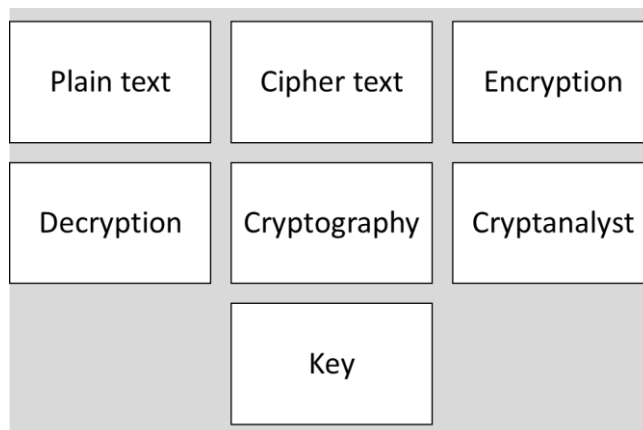


Figure 1 Basics of Network Security

14.1 Basics of Network security terms

A mono alphabetic substitution is a type of keyword encryption. The letter matchings of the cypher alphabet to the plain alphabet are determined by a keyword that serves as the key. The cypher alphabet is formed with the keyword matching to A, B, C, and so on until the keyword is exhausted, at which point the remaining cipher text letters are utilised in alphabetical sequence, omitting those previously used in the key.

Encryption

The first line of input includes the keyword you want to use. The string you must encrypt is on the second line of input.

Plain text : A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
Encrypted : K R Y P T O S A B C D E F G H I J L M N Q U V W X Z

All As become Ks, all Bs become Rs, and so on when the keyword is KRYPTOS. Using the keyword "kryptos" to encrypt the message "knowledge is power."

Encrypting the message: Knowledge is Power

Encoded message: IlmWjbaEbgqNmWbp

Few points to be noted in this method:

- All of the messages are written in capital letters.
- Although you can use whitespace, special characters, and digits in your keyword, they will not be considered.
- Whitespace, special characters, and integers are unaffected by the encryption process.

Decryption

To decode the message, compare the provided message's location in the encrypted text to the plain text.

Plain text : A B C D E F G H I J K L M N O P Q R S T U V W X Y Z
Encrypted : K R Y P T O S A B C D E F G H I J L M N Q U V W X Z

Message: PTYBIATLEP

Deciphered Text : DECIPHERED

Now, how we generate the deciphered string?

We search for 'P' in Encrypted Text and compare its position with plain text letter and generate that letter. So 'P' becomes 'D', 'T' becomes 'E', 'Y' becomes 'C' and so on.

Cryptography

Cryptography is a method of safeguarding information and communications by encoding it in a way that only the people who need to know can interpret and interpret it. As a result, unwanted access to information is prevented. The word "crypt" stands for "hidden," and the suffix "graphy" stands for "writing." The procedures used to safeguard information in cryptography are derived from mathematical principles and a set of rule-based calculations known as algorithms that change signals in ways that make them difficult to decipher. These algorithms are used to generate cryptographic keys, digitally sign documents, verify data privacy, browse the internet, and secure sensitive transactions like as credit card and debit card transactions.

Techniques used For Cryptography:

In today's computer era, cryptography is generally connected with the process of converting plain text to cypher text, which is text encoded in such a way that only the intended receiver of the information can decode it, a process known as encryption. Decryption refers to the process of converting encrypted text to plain text.

Features of Cryptography are as follows:

Confidentiality: Only the individual for whom the information is meant has access to it, and no one else has access to it.

Integrity: Information cannot be changed in storage or in transit between the sender and the intended receiver without being noticed.

Non-repudiation: The information creator/sender cannot dispute that he or she intends to convey information at a later time.

Authentication: The sender's and receiver's identities have been verified. The information's destination/origin is also confirmed.

Types of cryptography

There are three forms of cryptography in general:

1. Symmetric Key Cryptography

It is an encryption scheme in which the sender and receiver of a message encrypt and decode messages using a single shared key. Symmetric Key Systems are quicker and easier to use, but they have the drawback of requiring the sender and receiver to exchange keys in a safe way. Data Encryption Technology is the most widely used symmetric key encryption system (DES).

2. Hash functions

This algorithm does not make use of any keys. A hash value with a defined length is computed based on the plain text, making it difficult to reconstruct the plain text's contents. Hash functions are used by several operating systems to encrypt passwords.

3. Asymmetric key cryptography

A pair of keys is used to encrypt and decode data in this system. Encryption is done using a public key, while decryption is done with a private key. The terms "public key" and "private key" are not interchangeable. Even though everyone knows the public key, the intended receiver can only decode it since he is the only one who knows the private key.

Cryptanalyst

The decoding and examination of codes, cyphers, or encrypted text is known as cryptanalysis. Cryptanalysis is a method of searching for algorithm flaws and breaking into cryptography or information security systems using mathematical formulae.

The following are examples of cryptanalysis attacks:

Known-Plaintext Analysis (KPA): With knowing partial plaintext, the attacker decrypts ciphertexts.

Chosen-Plaintext Analysis (CPA): Using the same mathematical methodology, the attacker creates ciphertext that matches any picked plaintext.

Ciphertext-Only Analysis (COA): The attacker employs ciphertext collections that are well-known.

Man-in-the-Middle (MITM) Attack: When two parties utilise message or key sharing to communicate through a channel that appears to be secure but is really compromised, an attack occurs. This technique is used by the attacker to intercept messages as they move across the communications channel. MITM attacks are avoided by using hash functions.

Adaptive Chosen-Plaintext Attack (ACPA): This attack employs selected plaintext and ciphertext depending on data learnt from previous encryptions, similar to a CPA.

14.2 Network Security Issues

Name	Definition	How could it affect you?
Hacking	Use a computer to gain unauthorized access to data in a system	A serious crime, leading to the theft of identifying information or the shutdown of online services. Even those who are not the direct victims of hackers can be affected by computer hacking.
War Driving	The act of searching for Wi-Fi wireless networks by a person in a moving vehicle, using a portable computer, smartphone or personal digital assistant	Spotting a vulnerable network via war driving, infiltrating it to steal online banking log in information and then using it to transfer or withdraw funds from the company's accounts.
Computer Viruses	A piece of code that is capable of copying itself and typically has a detrimental effect, such as corrupting the system or destroying data	Cause your computer to slow down and can also result in the loss of important files.
Computer Worm	A standalone malware computer program that replicates itself in order to spread to other computers	Exploit operating system vulnerabilities. Worms typically cause harm to their host networks by consuming bandwidth and overloading web servers.
Trojan Horse	A program designed to breach the security of a computer system while ostensibly performing some innocuous function.	Reforming background tasks such as giving access to your computer or sending personal information to other computers. Trojan horses are one of the most common
		methods a criminal uses to infect your computer and collect personal information from your computer
Denial of Service Attacks	An attempt to make a machine or network resource unavailable to its intended users.	User or organization is deprived of the services of a resource they would normally expect to have. Typically, the loss of service is the inability of a particular network service, such as e-mail, to be available or the temporary loss of all network connectivity
Data or Program Alteration	When your computer is hacked and your programing and data is altered and transformed.	When you were to log on to your computer after this has taken place your data and programing would be out of order and disorganized.

14.3 Security Goals

Data is extremely vulnerable to assaults during transmission. To fulfil his malicious goals, an attacker can target the communication channel, collect the data, and read or re-insert a bogus message. Network security is concerned not only with the security of the computers at each end of the communication chain, but also with the security of the entire network.

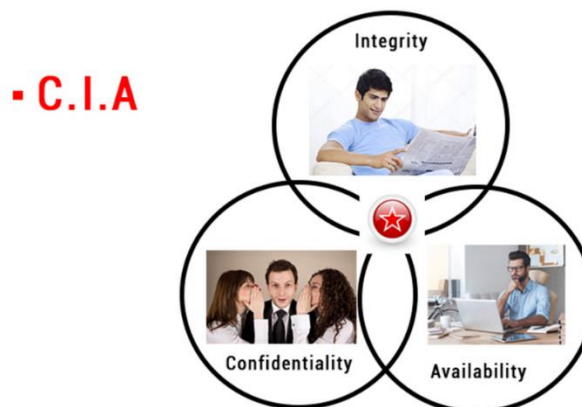


Figure 2 Security Goals

Protecting the usefulness, dependability, integrity, and safety of a network and its data is what network security is all about. Effective network security prevents a wide range of dangers from infiltrating and propagating throughout a network.

Confidentiality, Integrity, and Availability are the three main goals of network security as seen in Figure 2. The CIA triangle is widely used to depict these three pillars of network security as seen in Figure 3.

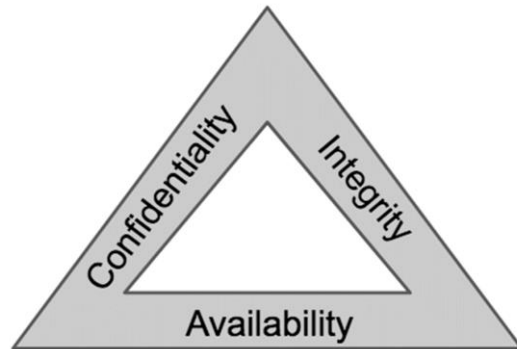


Figure 3 CIA Triangle

Confidentiality: The purpose of secrecy is to keep sensitive company information safe from prying eyes. The confidentiality component of network security ensures that data is only accessible to those who are authorised to see it.

Integrity: This aim entails ensuring and maintaining data correctness and consistency. The purpose of integrity is to ensure that data is accurate and not tampered with by unauthorised individuals.

Availability: The purpose of availability in Network Security is to ensure that data, network resources, and services are always available to authorised users when they need them.

14.4 Security Services

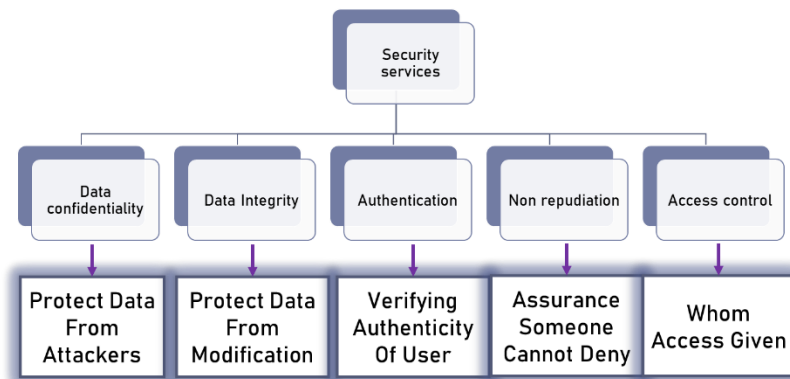


Figure 4 Security Services

Data confidentiality

The safeguarding of data from unlawful disclosure.

Connection Confidentiality: The protection of all user data on a connection. **Connectionless Confidentiality:** The protection of all user data in a single data block.

Selective-Field Confidentiality: The confidentiality of selected fields within the user Data on a connection or in a single data block.

Traffic Flow Confidentiality: The protection of the information that might be Derived from observation of traffic flows.

Data Integrity

The assurance that data received is identical to what was sent by an entity.authorised entity (i.e., no changes, additions, or deletions)or a rerun).

Authentication

The act of verifying a claim, such as a computer system user's identity, is known as authentication. Authentication is the process of validating a person's or thing's identity, as opposed to identification, which is the act of indicating that person's or thing's identity.

Non-repudiation

The term "non-repudiation" refers to a circumstance in which the author of a statement is unable to effectively contest the statement's authorship or the validity of an associated contract. When the legitimacy of a signature is being questioned in court, the word is frequently used. The genuineness is "repudiated" in this situation.

Access control

Individuals are authenticated and authorised to access the information they are permitted to see and use through access restrictions.

14.5 Approaches to network security

The Castillo of San Felipe de Barajas in Cartagena is the Spanish colonies' largest and most powerful stronghold. The fort, which was built between 1639 and 1657 and subsequently expanded significantly in 1762, defended the "portal to the new world" against several assault and was never captured.A fleet of Spanish ships patrolling in concentric "rings of defence" would distract or distract any opponent approaching the stronghold, preventing any threat to the citadel. If the adversary were to battle through these rings, the opponent would be protected from cannon fire and ground attack by the high, thick walls, while a sequence of batteries and parapets placed to cover each other would eliminate attack.

The fortress's vast maze of tunnels deep within its core is perhaps the most fascinating aspect. If the fort was ever taken, the defenders would escape into this maze, reorganise, and retake the citadel from within.These multiple levels of defence, though not a perfect parallel, highlight three unique approaches to security. Preventing a danger from forming in the first place, particularly through addressing its underlying causes, is one security strategy.

When a hazard cannot be avoided, security as protection seeks to mitigate, if not completely eradicate, the hazard. However, if we are unable to totally shield ourselves from the danger, security as resilience analyses our capacity to "bounce back" and change the manner in which it impacts our social systems - our capacity to adapt to genuine dangers.

What they target: the fundamental causes of threats, the threats themselves, or the qualities of the referent that is endangered is the essential component that distinguishes the three methods. However, as shown in the table below, the three techniques can be separated in a variety of additional ways.

Table 1 Approaches to network security

Security Approach:	Security as Prevention	Security as Control (Protection)	Security as Resilience
Definition	This strategy aims to prevent dangers from forming in the first place by addressing the root issues that lead to their emergence.	This strategy aims to manage, defend against, or remove a clear threat.	Security as resilience focuses on the ability of social systems to "bounce back" and recover from shocks when dangers cannot be managed or removed.It is about a society's flexibility and adaptation, as well as its rigidities

			and how to lessen its vulnerability to disturbance and collapse.
Focus/Target of the Approach	Threats and their underlying causes	The dangers they pose	The endangered referent's capacity to recover or adapt in the face of threats, in particular.
Ontology	The combination of broad underlying structural reasons and proximate factors that mobilise these foundations into tangible threats produces threats.	A Newtonian-mechanistic cosmos with a reasonable number of variables and reasonably straightforward and predictable causality. Problems may be broken down and isolated into their constituent elements.	A complicated cosmos with too many variables to count, emergent features, non-linear causality, phase shifts, limited predictability, and unexpected shocks. Threats originate from the interplay of several variables and are not reducible to individual components.
Threat Type	Economic disparity, relative deprivation, poor governance, and political exclusion as underlying causes of civil war are best suited.	Best suited to threats that can be reduced to specific persons, organisations, and events and then defended against or removed, such as a rival state developing a nuclear programme or a traditional military threat.	Complex adaptive systems that can self-organize (not reducible to individual players) and adapt to defensive measures while eluding elimination, such as the drug trade, which has confounded four decades of drug prohibition, are most adapted to threats.
Example & Approach: The Drug Trade	Reduced demand.	The drug war targets prominent drug lords and organisations with military strikes, with the ultimate objective of eradicating the drug trade by force; crop eradication; and drug interdiction.	Harm reduction measures that lessen the impacts of drug use; decriminalization of some drugs to ease the enforcement burden; and enforcement measures that target only the most violent traffickers.
Example & Approach: Terrorism	Addresses both the frustrations that drive terrorist acts and the societal factors that attract recruits.	Drone strikes and other counter-terrorism tactics are used to locate, apprehend, or murder terrorist leaders, while intelligence and enforcement efforts	Decoupling, contingency planning, and redundancy ensure that our social systems (trade, transport, communications, energy, and so on)

		disrupt current terrorist plots.	can bounce back after an assault; heightened security measures do not overwhelm social systems with complexity, interruptions, resource needs, and rigidities.
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14.6 Difference between private key and public key

The difference between private key and public key are shown in Table 2.

Table 2 Private key vs Public key

Private Key	Public Key
Private key is faster than public key.	It is slower than private key.
In this, the same key (secret key) and algorithm is used to encrypt and decrypt the message.	In public key cryptography, two keys are used, one key is used for encryption and while the other is used for decryption.
In private key cryptography, the key is kept as a secret.	In public key cryptography, one of the two keys is kept as a secret.
Private key is Symmetrical because there is only one key that is called secret key.	Public key is Asymmetrical because there are two types of key: private and public key.
In this cryptography, sender and receiver need to share the same key.	In this cryptography, sender and receiver does not need to share the same key.
In this cryptography, the key is private.	In this cryptography, public key can be public and private key is private.

Summary

- Data on the network is not confidential, thus it must be kept safe from unauthorized users who may be lurking behind networked workstations.
- The malevolent goals might include knocking down network servers, utilizing people's sensitive information such as credit card details for illegal activities, and disrupting large corporations' websites. As a result, it aims to protect data and prevent eavesdroppers from listening in and stealing it. User data on a computer is also safeguarded by granting password-protected access to data and resources, allowing only authorised users to access them. Identifying criminals and foiling their attempts to inflict damage to the network and other resources are also important security considerations.
- Authentication entails validating the antecedents of the person who has requested remote machine services or access, either physically or over e-mail, before permitting him or her to do so. The process of authenticating a person's identification to a distant system is known as authentication.
- The authenticity of a message received by a remote machine is referred to as integrity. In other words, the message transmitted by the source machine is exactly the same as it was

before. The cyclic redundancy coding approach will not suffice in this scenario, as attackers in the system or communication channel may modify the message on purpose.

- Confidentiality: It assures that no one can read the message while it is in transit. This demands the use of encryption techniques in the future.
- With the use of encryption and decryption techniques, the communication is encrypted at the sender end and decoded at the receiving end to ensure anonymity. The secret key and public key procedures are the two options, each with its own set of benefits and drawbacks.
- Transposition and substitution In traditional cryptography, there are two types of cyphers. The encryption mechanism handles parts of the message differently in substitution and transposition.

Keywords

Cipher text: The encrypted message formed by applying the method to the plaintext message using the secret key is known as ciphertext.

IP-spoofing: IP spoofing, like honey pots, entails a computer successfully impersonating a trusted server/resource and intercepting data packets.

Maliciously: Maliciously programmed websites generate charitable webpages that allow users to make donations while also collecting important personal information.

Packet Sniffers: Packet sniffers are devices that intercept data streams over a network in order to gather sensitive information such as usernames, passwords, and credit card numbers.

Password Attacks: A 'Password Attack' includes a number of techniques used by hackers to steal passwords.

Phishing: Emails with titles such as, "URGENT: Update Account Status" are all attempts by a spammer to "phish" the account details.

Plain text: It is the text message to be transmitted on which an algorithm is applied.

Private key: The key of a key pair, which is used to create a digital signature. It can be used to sign a message that only the corresponding public key can verify

Public key: It is the key of a key pair that is used to verify a digital signature. Key pair consists of private and public key.

Secret Key: They constitute a part of algorithm for encryption and decryption of the message.

Self Assessment

1. In cryptography, what is cipher?
 - A. algorithm for performing encryption and decryption
 - B. encrypted message
 - C. both algorithm for performing encryption and decryption and encrypted message
 - D. decrypted message
2. In asymmetric key cryptography, the private key is kept by _____.
 - A. sender
 - B. receiver
 - C. sender and receiver
 - D. all the connected devices to the network

3. Cryptanalysis is used _____
- A. to find some insecurity in a cryptographic scheme
 - B. to increase the speed
 - C. to encrypt the data
 - D. to make new ciphers
4. The process of transforming plain text into unreadable text.
- A. Decryption
 - B. Encryption
 - C. Network Security
 - D. Information Hiding
5. A process of making the encrypted text readable again.
- A. Decryption
 - B. Encryption
 - C. Network Security
 - D. Information Hiding
6. Public-key cryptography is also known as ?
- A. asymmetric cryptography
 - B. symmetric cryptography
 - C. Both A and B
 - D. None of the above
7. Which of the following keys are known only to the owner?
- A. public key
 - B. protected key
 - C. private key
 - D. unique key
8. PKI stands for?
- A. public key infrastructure
 - B. private key infrastructure
 - C. public key instance
 - D. private key instance
9. In which of the following, a person is constantly followed/chased by another person or group of several peoples?
- A. Phishing
 - B. Bulling
 - C. Stalking
 - D. Identity theft
10. Which of the following refers to the violation of the principle if a computer is no more accessible?

- A. Access control
- B. Confidentiality
- C. Availability
- D. All of the above

11. The term "TCP/IP" stands for _____

- A. Transmission Contribution protocol/ internet protocol
- B. Transmission Control Protocol/ internet protocol
- C. Transaction Control protocol/ internet protocol
- D. Transmission Control Protocol/ internet protocol

12. In the computer networks, the encryption techniques are primarily used for improving the _____

- A. Security
- B. Performance
- C. Reliability
- D. Longevity

Answers for Self Assessment

- | | | | | |
|-------|-------|------|------|-------|
| 1. A | 2. B | 3. A | 4. B | 5. A |
| 6. A | 7. C | 8. A | 9. C | 10. C |
| 11. B | 12. A | | | |

Review Questions

1. What are different criteria to keep information private when it is sent over a public network?
2. How does the encryption affect performance of network?
3. There are certain information bases on the Internet that need to be prevented by undesirable person to get. How can undesirable person be kept from accessing this?
4. How do we keep our own and other people's computers safe from hackers? Explain with the help of a hypothetical situation.
5. What is a Cipher? Why are cipher used for large messages?
6. Describe briefly two kinds of security attacks, which can be directed against an Internet connected computer system
7. What is the difference between secret key and public key encryption?
8. What is cryptography? What are the benefits of using this technique?
9. What do you mean by substitution and transposition ciphers? Differentiate between the two.

Further Readings

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